

Analytical Problems.
CO2 Assessment Tool-1

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Problem Interpretation :

Given the sorted dataset :

[3, 7, 15, 21, 27, 33, 39]

We need to find the element 33 using Binary Search.

Binary search is suitable because the list is already arranged in ascending order. Instead of checking every element one by one. Binary search checks the middle element and eliminates half of the remaining search area after each comparison.

2) Step-by-step Binary Search.

step	search interval	middle Element
1	[3, 7, 15, 21, 27, 33, 39]	21
2	[27, 33, 39]	33

step 1

[3, 7, 15, 21, 27, 33, 39]

↑
middle

Compare :

$33 > 21$

Therefore, 33 must be in the right half.

The new search interval becomes :

[27, 33, 39]

Step 2:

[27, 33, 37]

↑

The middle element is 33.

Compare

33 = 33

Therefore found.

Result:

Element 33 found at index 5

Using 1-based position, 33 is at position 6.

Time Complexity:

Binary search eliminates approximately half of elements after every comparison.

For n elements:

* Best case: $O(1)$ - target is found at the first middle element

* Avg case: $O(\log n)$

* Worst case: $O(\log n)$

For this list of 7 elements the target was found after only 2 comparisons.

∴ The space complexity of the iterative Binary search algorithm is $O(1)$.